

Why $W_2(\delta_0, \delta_1) = 1$

Let

$$\mu = \delta_0, \quad \nu = \delta_1.$$

By definition,

$$W_2(\mu, \nu) = \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int |x - y|^2 d\gamma(x, y) \right)^{1/2}.$$

Since all mass starts at 0 and must end at 1, the only possible transport plan is

$$\gamma = \delta_{(0,1)}.$$

Thus

$$\int |x - y|^2 d\gamma(x, y) = |0 - 1|^2 = 1.$$

Therefore,

$$W_2^2(\delta_0, \delta_1) = 1, \quad W_2(\delta_0, \delta_1) = \sqrt{1} = 1.$$

More generally,

$$W_p(\delta_a, \delta_b) = |a - b|.$$